

JOMALA, Finland

WMO: 029710

Lat: 60.150N

Lon: 19.867E

Elev: 33

StdP: 14.68

Time Zone: 2.00 (EUE)

Period: 97-06

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.4	7.5	-3.8	4.5	0.7	3.6	6.6	8.4	25.2	33.0	21.9	31.6	0.6	80	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.0	77.0	65.1	73.9	63.3	70.9	62.0	67.4	74.2	65.3	71.2	63.4	68.7	5.5	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.7	91.6	70.4	62.9	85.8	68.8	61.0	80.2	66.5	31.8	74.1	30.2	71.3	28.8	69.0	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.1	15.4	13.2	DB	-6.5	80.6	6.9	4.7	-11.4	84.0	-15.4	86.7	-19.3	89.4	-24.3	92.8	(4)
(5)			WB	-7.0	68.7	6.7	3.0	-11.8	70.8	-15.7	72.6	-19.4	74.3	-24.3	76.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	42.7	28.3	26.9	29.0	39.0	46.9	55.0	61.9	59.3	52.8	43.7	36.7	31.6		
	DBStd	13.71	9.14	8.59	7.93	5.59	5.27	4.75	4.19	5.13	6.06	6.90	6.73	6.98		
	HDD50	3664	674	646	651	330	124	12	0	1	43	214	398	571		
	HDD65	8175	1139	1066	1116	779	562	301	115	186	367	660	848	1036		
	CDD50	994	0	0	0	1	26	161	369	291	127	19	0	0		
	CDD65	31	0	0	0	0	0	0	19	11	1	0	0	0		
	CDH74	255	0	0	0	0	1	6	146	102	0	0	0	0		
	CDH80	21	0	0	0	0	0	0	13	8	0	0	0	0		
	WSAvg	5.0	5.5	5.6	5.3	5.2	5.4	5.1	4.4	4.0	4.3	4.9	5.1	5.6		
	PrecAvg	20.9	1.5	1.0	1.0	1.1	1.1	1.3	2.1	2.7	2.6	2.2	2.5	1.8		
(15) Precipitation	PrecMax	30.8	3.5	3.3	2.2	2.3	2.4	3.5	5.5	4.9	6.8	5.9	4.9	3.6		
	PrecMin	14.6	0.4	0.2	0.0	0.0	0.2	0.1	0.7	0.5	0.7	0.9	0.7	0.3		
	PrecStd	3.9	0.8	0.7	0.5	0.6	0.7	0.8	1.3	1.2	1.3	1.3	1.0	0.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.4	43.5	49.5	62.1	70.1	74.9	82.2	81.1	71.6	59.0	50.7	45.9		
		MCWB	40.6	39.3	41.6	51.2	55.2	60.8	70.1	65.6	62.7	55.3	47.0	44.2		
	2%	DB	40.4	41.3	44.6	56.2	64.9	71.7	77.9	76.9	67.1	57.2	48.6	44.2		
		MCWB	38.4	38.7	39.3	47.8	53.7	59.0	65.9	65.0	59.8	54.7	46.5	42.7		
	5%	DB	39.0	39.0	41.2	52.7	60.2	68.1	74.9	73.1	64.2	55.4	47.4	43.3		
		MCWB	37.3	36.9	37.2	45.2	51.0	58.4	64.1	63.9	58.4	53.1	45.7	41.8		
	10%	DB	37.7	37.3	38.5	49.5	57.5	65.2	71.9	69.5	62.3	53.3	46.1	41.1		
		MCWB	36.5	35.6	35.5	43.7	49.6	57.3	62.7	62.2	57.8	51.7	44.7	39.6		
	0.4%	WB	41.2	40.7	42.7	51.3	58.2	63.6	70.2	69.6	63.4	56.7	48.2	44.9		
		MCDB	41.9	42.2	48.8	59.5	67.8	70.5	79.4	76.6	69.1	57.8	49.2	45.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	39.0	38.8	39.8	48.7	54.6	60.7	68.1	67.3	61.0	54.9	47.1	43.5		
		MCDB	39.7	40.6	44.0	54.9	61.6	68.0	75.6	73.9	65.3	56.6	48.2	44.2		
	5%	WB	37.8	37.4	37.7	46.6	52.3	59.3	66.2	64.7	59.4	53.5	46.1	42.0		
		MCDB	38.6	38.9	40.5	51.5	59.2	66.2	72.1	71.0	63.1	55.1	47.3	43.0		
	10%	WB	36.8	35.8	36.1	44.0	50.4	57.8	64.6	62.8	58.1	51.8	44.8	39.9		
		MCDB	37.6	37.2	38.2	48.3	56.1	63.8	69.8	67.8	61.4	53.1	46.0	40.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	10.1	12.6	14.3	15.5	15.0	15.0	15.9	14.6	10.2	7.2	7.0		
		MCDBR	7.1	7.7	15.0	21.4	21.8	21.6	22.3	21.7	16.4	10.5	7.8	5.5		
		MCWBR	6.6	6.0	10.7	13.6	12.8	11.4	11.8	11.3	10.5	8.2	7.2	5.2		
		MCDBR	6.5	7.1	13.8	19.0	18.7	17.8	17.4	19.6	14.6	8.5	6.9	5.4		
	5% WB	MCWBR	6.4	5.7	10.4	12.8	11.8	10.4	10.2	11.6	10.0	7.1	6.8	5.3		
(40) Clear-Sky Solar Irradiance	taub		0.276	0.312	0.332	0.370	0.367	0.358	0.387	0.377	0.347	0.333	0.301	0.389		
	taud		2.187	2.259	2.321	2.337	2.422	2.490	2.432	2.447	2.517	2.418	2.228	2.660		
	Ebn at Noon		155	216	252	263	273	277	265	259	249	209	142	110		
	Edh at Noon		13	22	28	33	32	31	32	30	23	19	13	9		
(44) All-Sky Solar Radiation	RadAvg		77	257	741	1271	1728	1921	1789	1394	863	369	104	49		
	RadStd		10	48	81	116	143	123	166	119	92	56	23	7		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	-193	N/A	N/A	N/A

Nomenclature: See separate page